



4 Earthquakes

Much of what is known about earthquakes follows from study of the motion of the ground.

Charles Richter, *Elementary Seismology*, 1958

4.1 Introduction

Seismology deals with the generation and propagation of seismic waves. Our initial focus has been on the propagation of seismic waves and how they can be used to study the interior of the earth. We now turn to the generation of seismic waves and how they are used to study earthquakes. This association is so strong that seismology is sometimes viewed as the science of earthquakes, rather than of elastic waves in the earth. Both definitions are used, but the latter has become more common because seismology is the primary tool used to investigate earth structure as well as earthquakes, whereas techniques other than elastic waves are also used to investigate earthquakes.

Earthquakes almost invariably occur on *faults*, surfaces in the earth on which one side moves with respect to the other. Typically, earthquakes occur on faults previously identified by geological mapping, which shows that motion across the fault has occurred in the past. Earthquakes that occur on land and close enough to the surface often leave visible ground breakage along the fault. For example, earthquakes occur along the San Andreas fault, which can be seen cutting across California for great distances (Fig. 4.1-1). One of these, the famous 1906 magnitude 7.8 San Francisco earthquake on the San Andreas fault was one of the first US earthquakes to be studied carefully. Contemporary accounts showed that several meters of relative motion occurred along several hundred kilometers of the San Andreas fault (Fig. 4.1-2).

The earthquake and the resultant fires did such damage (Fig. 1.2-10) that a study commission was formed. As part of the investigation, H. Reid proposed the *elastic rebound* theory of earthquakes on a fault. In this model, materials at distance on opposite sides of the fault move relative to each other, but friction on the fault “locks” it and prevents the sides from slipping (Fig. 4.1-3). Eventually the strain accumulated in the rock is more than the rocks on the fault can withstand, and the fault



Fig. 4.1-1 Aerial photograph of the San Andreas fault in the Carrizo Plain in California, seen from the south. Note the displacement of stream gullies as the Pacific plate (*near side*) has moved to the left (northwest) relative to North America. (Copyright John S. Shelton.)

slips, resulting in an earthquake. The motion illustrated in this cartoon by an offset fence can sometimes be seen after earthquakes using other linear features, including rows of trees, railroad tracks, or roads (Fig. 4.1-4).

The elastic rebound idea was a major conceptual breakthrough, because the faulting seen at the surface had been previously regarded as an incidental side effect of an earthquake, rather than its cause. Subsequently, earthquake studies have been widely pursued for several reasons. One is to understand the large-scale geological processes causing earthquakes. It

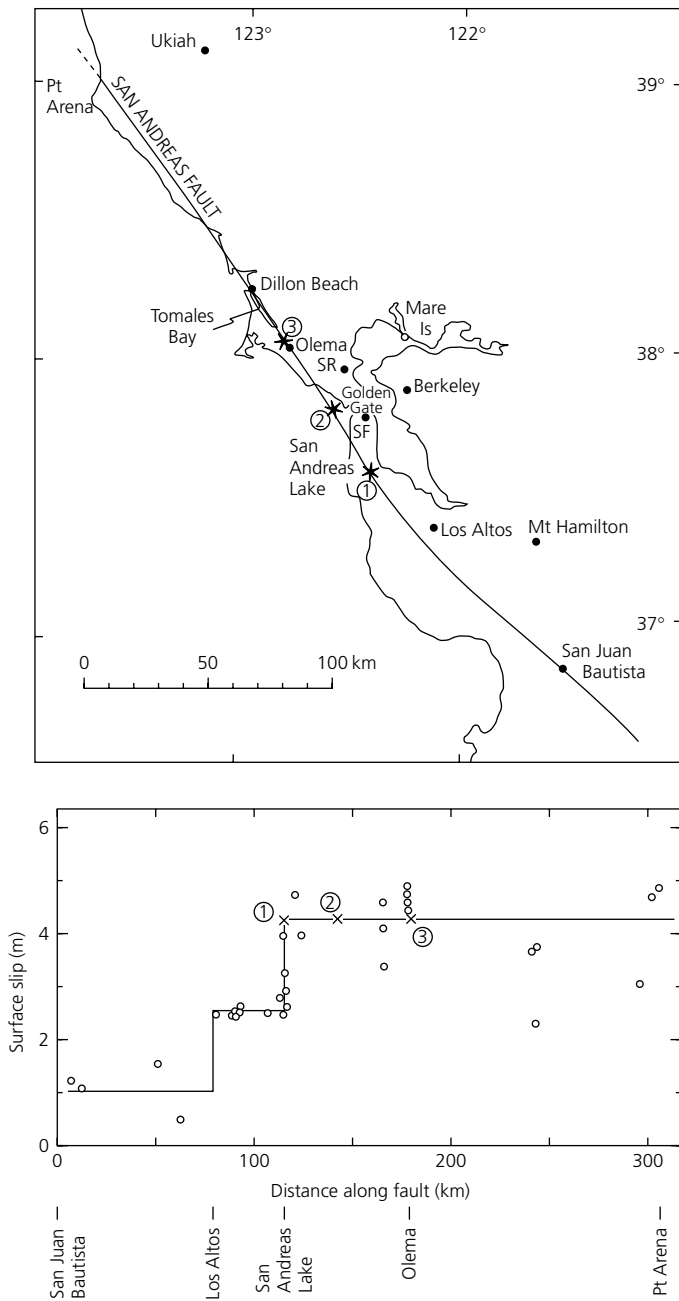


Fig. 4.1-2 Map of the portion of the San Andreas fault that slipped in the 1906 San Francisco earthquake (*top*) and the amount of surface slip reported at various points along it (*bottom*). This slip is the distance by which the earthquake displaced originally adjacent features on opposite sides of the fault. (Boore, 1977. © Seismological Society of America. All rights reserved.)

turns out that earthquakes largely reflect the motions of lithospheric plates, and so provide valuable information about how and why plates move. For example, earthquakes on the San Andreas fault result from the steady motion between the

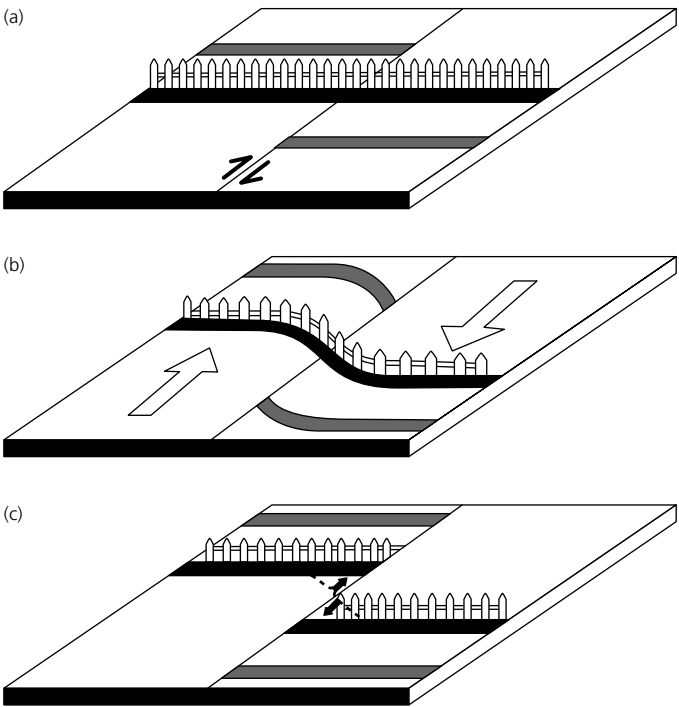


Fig. 4.1-3 The elastic rebound model of earthquakes assumes that between earthquakes, material on the two sides of a fault undergoes relative motion. Because the fault is locked, features across it that were linear at time (a), such as a fence, are slowly deformed with time (b). Finally the strain becomes so great that the fault breaks in an earthquake, offsetting the features (time c). (Courtesy of S. Wesnousky.)



Fig. 4.1-4 Displacement of crop rows resulting from slip along the Imperial fault, El Centro, California, on October 15, 1979. (Courtesy of the National Geophysical Data Center.)

North American and Pacific plates (Fig. 5.2-3). A second reason is to understand the fundamental physics of earthquake faulting. There are many unanswered questions about how and when faults break, even for earthquakes that occur near

the earth's surface, where data are relatively easy to gather. These issues are important for society because, as discussed in Chapter 1, knowledge of where and when earthquakes are likely, and of the expected ground motion during them, can help mitigate the risk they pose.

The largest earthquakes typically occur at plate boundaries. Using elastic rebound theory, we think of them as reflecting the most dramatic part of a process called the *seismic cycle*, which takes place on segments of the plate boundary over hundreds to thousands of years. During the *interseismic* stage, which makes up most of the cycle, steady motion occurs away from the fault but the fault itself is “locked,” although some aseismic creep can also occur on it. Immediately prior to rupture there is the *preseismic* stage that can be associated with small earthquakes (foreshocks) or other possible precursory effects. The earthquake itself marks the *coseismic* phase during which rapid motion on the fault generates seismic waves. During these few seconds, meters of slip on the fault “catch up” with the few mm/yr of motion that occurred over hundreds of years away from the fault. Finally, a *postseismic* phase occurs after the earthquake, and aftershocks and transient afterslip occur for a period of years before the fault settles into its steady interseismic behavior again.

Studying this cycle is difficult because it extends for hundreds of years, so we do not have observations of it in any one place. Instead, we have observations from different places, which we assume can be combined to give a complete view of the process. It is far from clear how good that view is and how well our models represent its complexity. As a result, earthquake physics remains an active research area that integrates a variety of techniques. Most faults are identified from the earthquakes on them, and seismology is the primary tool used to study the motion during the earthquakes and infer the long-term nature of motion on the faults. Moreover, because earthquakes are such dramatic events, historical records of earthquakes are often available and provide data on the earthquake cycle for a given fault or fault segment. Field studies, both on land and under water, also provide information about the location, geometry, and history of faults. Geodetic measurements are used to study ground deformation before, during, and after earthquakes, and thus the processes associated with fault locking and afterslip. For oceanic regions and deep earthquakes, where geodetic and geological observations are not available, almost all of what we know about the earthquakes themselves comes from seismology. The results for individual earthquakes are then combined and integrated with those from other techniques, as discussed in the next chapter, to better understand how earthquakes in a given region reflect the large-scale tectonic processes that cause them.

Of these approaches, our primary focus in this book is the information that seismology provides about earthquakes. The arrival time of seismic waves at seismometers at different sites is first used to find the location of an earthquake, known as the *focus*, or *hypocenter*, using techniques discussed in Chapter 7. Next, as discussed in this chapter, the amplitudes and shapes of

the radiated seismic waves are used to study the size of the earthquake, the geometry of the fault on which it occurred, and the direction and amount of slip. We introduce these techniques and discuss their applications, while leaving their derivation and details for more advanced treatments listed at the chapter's end.

It is worth bearing in mind that learning about earthquake faulting from the seismic waves that are generated is an inverse problem, like learning about earth structure from seismic waves. As discussed in Section 1.1.2, this means that studying seismic waves alone is limited in what it can tell about the earthquake process. We will see that the seismic waves radiated from an earthquake reflect the geometry of the fault and the motion on it, and so can give an excellent picture of the *kinematics* of faulting. However, they contain much less information about the actual physics, or *dynamics*, of faulting. In the next chapter, we discuss how seismological results are being combined with experimental and theoretical studies of rock friction and fracture to explore the physics of earthquakes.

4.2 Focal mechanisms

4.2.1 Fault geometry

To describe the geometry of a fault, we assume that the fault is a planar surface across which relative motion occurred during an earthquake. Geological observations of faults that reach the surface show that this is often approximately the case (Fig. 4.2-1), although complexities are common. Similarly, we will see that this assumption is usually (but not always)



Fig. 4.2-1 Fault cutting across a moraine near Crowley Lake, California. The land in front has dropped relative to the background. (Copyright John S. Shelton.)